

Artificial Intelligence for QA

AI types, LLMs, generative tools and responsible applications across the quality lifecycle.

AI, Machine Learning and Deep Learning

AI is the broad field. Machine Learning uses data to learn patterns; Deep Learning uses deep neural networks. Generative AI produces new content, and LLMs are models specialized in language.

Tools and uses

ChatGPT, Claude, Gemini, DeepSeek, Copilot and other tools have different capabilities and integrations. In QA they can support requirements analysis, scenario brainstorming, code, documentation and investigation, always with human review.

RAG and agents

RAG retrieves relevant information from a knowledge base and supplies it to the model to ground answers. Agents can combine models with tools and decision steps to execute more complex workflows.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

Good practices

Prioritize high-impact scenarios, keep data and environments controlled, record important decisions and review the approach periodically to avoid redundant or outdated coverage.

QA tip

Never send credentials, personal data, confidential code or client information to AI services without proper authorization and controls.